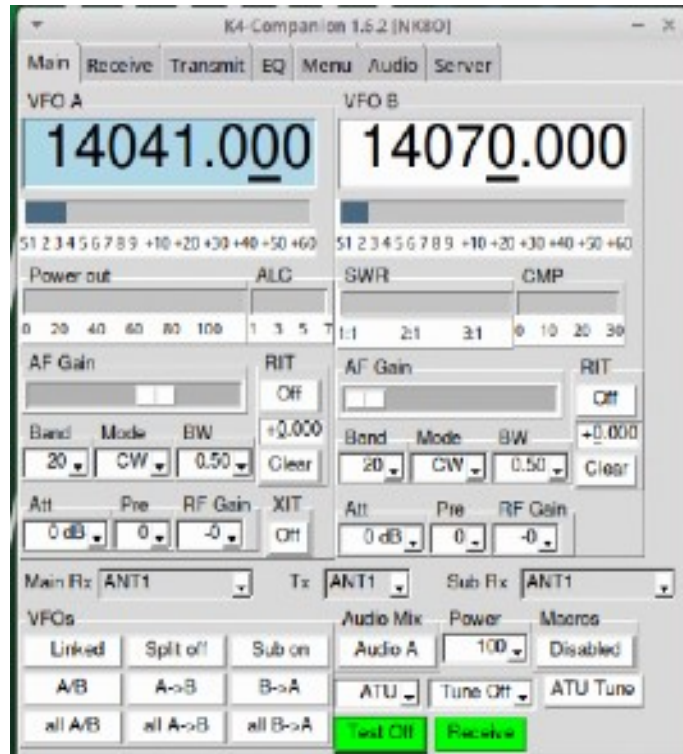


K4-Companion

An application for remote control of Elecraft the K4 series SDR transceiver



Developed by Dale Farnsworth, W7DA

Based on a simple utility originally devised by Charles Powell, NK8O

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The current version of K4 Companion is v1.6.2

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Introduction

K4 Companion is an application written in python3 that can remotely control an Elecraft K4 transceiver via TCP/IP. It currently controls the main K4 features and is very usable as is, but new features are being added all the time. K4 Companion is very configurable.

K4 Companion began life as a simple macro-sending program called K4Macro-Python, created by Charles Powell, NK8O. It has now grown far beyond a simple macro-sending program into a full-fledged remote control program for the K4.

Please send problem reports either: by sending an email, by entering an issue on github, or by making a pull request. Problem reports and suggestions are greatly appreciated.

Configuration can be modified a separate YAML file named, by default, k4companion.yaml. Custom configurations can be loaded with the *-config* option between the python executable and the desired configuration file, but it is no longer necessary to use a YAML file to run the default configuration.

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License

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#

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Windows Installer & Binaries

A Windows installer and Windows binary executable files are now available. The Windows installer places all items in the proper directory, creates a Start Menu entry, and it also will create a desktop shortcut if that option is selected. Double click the installer, then select the desired options. The user manual is also provided, found in %PROFILE%/k4companion/**You may need to disable Windows security briefly to download the installer.** If you are not present at your K4, a way to start your K4 remotely is also needed. Please refer to the K4 User Manual or any of several forums that address this function.

NOTE: The installer automatically places opus.dll, a mandatory dependency, in the %PROFILE%/k4companion directory. K4 Companion looks for opus.dll in the same directory (folder) as k4companion.exe. The Windows installer does this automatically. K4 Companion has been compiled under Windows 11, and tested on both Windows 11 and Windows 10. It is built on Python 3.14

If preferred, manual installation is possible. There are three files that are needed to run under the Windows operating system. These are:

- k4companion.exe
- opus.dll

The three k4companion files should be placed in a directory (folder) of your choice. These need to be placed in the same directory (folder). When starting the program, a splash screen appears as the program opens (Windows only).

There is a configuration file, k4companion.yaml, if you chose to make changes to the program's behavior. This fully editable with your favorite text editor. It is NOT necessary to have this file installed, and you may break things if you don't know what you are doing. If you have made significant sequential changes to the YAML file, it is a good idea to save the working copies with some kind of identifier, and you may wish to offer your changes to others by creating a fork of K4-Companion.

A note regarding Windows Defender: The Defender reaction will range from none at all, to identifying K4-Companion as a virus. If identified as a virus, Defender will quarantine the executable and ask if you want to delete it. Instructions regarding bypassing Windows Defender are found at <https://answers.microsoft.com/en-us/windows/forum/all/how-to-allow-an-app-through-windows-defender/caab364c-9e65-448e-8635-98721be847ba>

Use of K4 Companion outside your local network requires forwarding of port 9205, a VPN, or creating a SOCKS 5 proxy. Unlike the K4 to K4 connections, K4 Companion uses port **9205**. The **K4 to K4** connection is made via port 9204. An alternative to opening ports is to employ Tailscale or WireGuard to access your internal network, thereby eliminating open ports. <https://tailscale.com> Skip to page 7 for information on running K4 Companion.

Installation on Linux

One of the beauties of open source software is that it is typically cross-platform. Using Python3, the application is available on any platform that has Python3 available. It was developed on Linux, but it has now been successfully ported to both Mac OS, and Windows. Nonetheless, performance on Linux is slightly superior to the other platforms.

Here are the items one needs to take full advantage of K4 Companion and all its features:

1. You must have a way to turn your radio on remotely unless you are sitting at the operating position. This is detailed in the K4 user manual (pages 17 & 18) and requires the momentary closure and grounding of pin 8 to pin 5 on the DB15 ACC connector. Unfortunately the Wake On LAN feature is not available for the K4. Other details are beyond the scope of this document. There are many operators who simply leave the radio on at all times.
2. A computer capable of running a full installation of Python is needed. This has been tested with Xubuntu, Debian 12, Chromebook, various iterations of the Dell XPS series, and several Lenovo laptops, and even an Evolve III notebook, running a variety of Linux flavors.
3. Install portaudio (*sudo apt install portaudio19-dev* on Debian based distributions)
4. Install Python3. This is pre-installed on most Linux distributions. Please refer to readily available documentation for installation on other operating systems.
5. Install 'python-tk' (*sudo apt install python3-tk* on Debian based distributions)
6. Install 'pip3', required for the remaining dependencies (*sudo apt install python3-pip*)
7. Install 'pyyaml' using pip
8. Install 'opuslib' for audio support using pip
9. Install 'numpy', and 'pyaudio' with using (Some of these may be pre-installed) portaudio is required for pyaudio to install correctly
10. Install 'PySocks' with pip. This allows SOCKS 5 proxy connections
11. For access **outside your local network, open port 9205 with port forwarding on your router**, a VPN, or SOCKS 5 proxy. K4 to K4 connections are accomplished via port 9204. You may use Tailscale to eliminate the need for open ports. <https://tailscale.com>, WireGuard, or other VPN systems. Operating the utilities is beyond the scope of this document

Notes: the typical installation of Python dependencies is with the command *pip install xxxxx* however security consciousness has caused the need for override with a number of operating systems. If your OS complains about installing the extras, use the command *pip install --break-system-packages xxxxx* . It doesn't really break anything. It installs the package in the individual user's profile. I have done many such installs and it has never been a problem. Incidentally the *pip* command can be used to install '.whl' Python files. Chirp, the well known open source HT programming software is provided this way, and using the .whl file is an easy way to keep up with changes and updates. *pipx* is an alternative to using *pip* and it usually does not require additional arguments.

For MacOS instructions see the document, "K4-Companion MacOS installation" in the Github repository.

Running K4 Companion

There are several ways to run K4 Companion, depending on the operating system. To execute the script directly, it requires the file to be marked as executable. Under Linux and Mac OS, this can be done immediately after downloading the script. In a terminal window and using the command line, issue the command `chmod 755 k4companion`. No modifications of the Windows executable files are needed.

Methods for starting K4 Companion:

- Assuming the file has been made executable, from a Linux terminal window issue the command `./k4companion &` (enter) from the directory or folder where the script, as well as the configuration file are located.
- The alternative method, which requires no modifications nor marking files as executable is as follows: `python k4companion &` (enter).
- To start the Windows binary navigate to the file in the chosen folder, then either click on `k4companion.exe` or highlight and press 'enter'
- If the Windows installer has been employed, K4 Companion can be run from the Start Menu or a Desktop icon if this option was selected at the time of installation
- In all cases, access from outside your local network requires access in one way or another to port 9205

Linux users: If you create a shortcut or desktop launcher K4 Companion, it must contain the full path to the executable in order to find the configuration file.

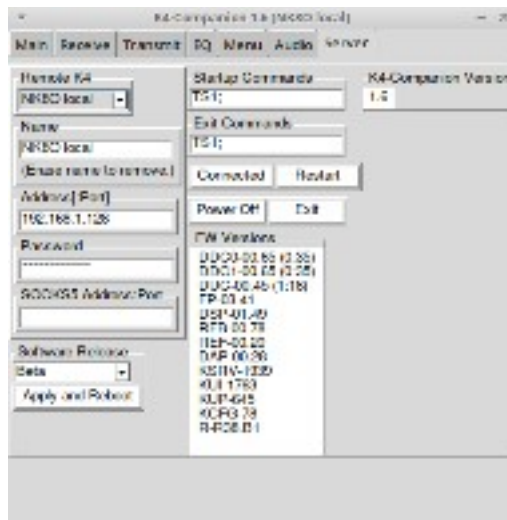
```
cwpowell@newXPS:~/bin$ ./k4companion &
```

```
cwpowell@newXPS:~/bin$ python3 k4companion &
```

Remote operation of K4 Companion can be achieved by one of several methods. Port forwarding through most home routers is possible. The default port is **9205**. An SSH 'tunnel' can be set, or a SOCK 5 proxy can be configured, or a VPN connection established. A service such as "Tailscale" may also be used. Tailscale overcomes some of the issues of port forwarding and CGNAT. It is free for personal use.

Simultaneous connection K4 to K4 on port 9204 and K4 Companion to K4 on port 9205 are not possible.

Configuration: Server tab



Server configuration is done through the 'Server' tab. Configuration parameters **must** be entered but K4 Companion will save these items and open after going back to the Main tab. The required items are:

- Specifying the 'Name' for the profile. Each new profile defaults to 'K4', but this can be changed when the profile is created or specified. Up to 10 profiles are possible with K4 Companion. In the 'Remote K4' pulldown after a profile is created, select the name for the connection. If only a single connection is created, K4 Companion will connect automatically. Otherwise the program opens to this page so the profile can be selected
- The address of the K4 to be controlled. This can either be the serial number of the radio in the form K4-SNxxxxx.local, or it can be the actual IP address of the K4. There are reports that using the actual IP address makes the connection more responsive, and the serial number setting may not work under Windows. Your experience may be different. For remote operation this will be either your external IP address, or a name you have created in a service such as DynDNS or other such services. The default is **port 9205**. Specifying the connection port is not necessary unless you have re-directed to a different port through your router
- The password to access your K4. For all remote control this **must** be set on the radio. The default password will NOT work. In addition, the number of connections to your K4 must also be specified. This can be set to one, and up to four connections allowed. No connections will be permitted if the parameter is set to zero or left at default.
- SOCKS 5 proxy is fully supported, although this is optional. The SOCKS 5 proxy address is entered in the 'SOCKS 5 Address:Port'. The default port is 1080 and need not be specified unless a different port is used. When operating from a single remote address, it is possible to "open" your router to accept only that address, making a proxy connection unnecessary. Nonetheless, a proxy connection offers greater flexibility in connection options
- Once information is entered, K4 Companion will attempt to connect. If the information is entered correctly and the server is running, a connection will be established. **Hint:** enter password information first, then the server address
- The Startup and Exit commands put the radio in Test Mode when K4 Companion opens and return it to Test Mode for safety when the program exits. These can be changed at your own risk!

Firmware upgrades or downgrades, and beta releases can be installed remotely. The column on the right shows what is actually installed and not available releases. Please check on the Elecraft site for information on the various releases and feature, or other information.

On the right hand side, the connection status is shown. '**Power Off**' turns off the K4. '**Exit**' will close K4 Companion without shutting down the radio. '**Restart**' can be used if configuration is changed or if there is a brief loss of connectivity, i.e., the Internet connection drops briefly or network congestion causes a problem. '**Connected**' reflects the connection status.

The version of K4 Companion is displayed at the upper right.

Operation and Controls

K4 Companion is logically organized with largely self-explanatory labels and functions. Each VFO can be linked, operated independently, swapped, placed into Split mode, Sub-receiver mode, or both simultaneously. A diversity receive selection is also available. Various other functions are controlled in the main tab, including preamps, attenuators, RF gain, ATU functions, and audio controls.

The active transmit window is shown in pastel pink, so it is possible to see at a glance which VFO will transmit when the radio is keyed. This prevents confusion when using Split or Sub modes.



In the image on the left, VFO A is active for transmitting, and on the right, VFO B is ready for transmit. Note below that the K4 is actually transmitting at 5 watts, the ALC is normal at 5, and the SWR is slightly greater than 1:1. The details available at a glance are the frequency of each VFO, the S-meter reading for each VFO (if both are active), power out either 0-10 watts or 0-110 watts, SWR, and for phone operation, compression. When transmitting, the active transmit VFO show dark blue

The scroll rate for each VFO can be set independently from 1 Hz to 10 kHz. The rate is determined by clicking on a segment of the VFO, then scrolling with touchpad, mouse, or arrow keys.

The VFO boxes also support **direct entry** Increment can be changed with the keyboard arrows ← → or mouse/touchpad.

The active scroll rate increment is underlined. The Up ↓ Down arrows on the keyboard can be used if there is a single click in the VFO box. Scrolling always occurs at the underlined increment, and this can be changed as noted above.



Pull-down menus shown in the images above are available for each VFO to set the following parameters:

- Band selection, 160 through 6 meters
- Mode selection, including all available modes for the K4 (does not include digital sub-modes)
- Bandwidth selection from 50 Hz to 5 kHz (for AM reception)
- RF attenuation in 3 dB steps up to -21 dB
- Preamp selection – none, 1, 2, or 3 (depending on the band)
- RF gain, in -10 dB increments down to -60 dB. Note: Elecraft recommends use of Attenuation rather than RF gain. Attenuation reduces the signal to the ADC. This is different from the K3

Each of these selections will “scroll” with the mouse or touchpad, in the same manner as the VFO controls.

The following controls are fairly self-explanatory, at least once you are familiar with K4 Companion.

When the application opens, it sets the K4 to Test Mode, as shown above. This button lights up yellow when engaged but turns green when off. Test Mode on startup prevents unintended transmission when setting up. **This can be disabled in the apps tab**, but it can be a useful feature. For safety, K4 Companion returns the radio to Test Mode on exit. Using Test Mode, it is also possible to check **ALC** and **compression** without actually transmitting.



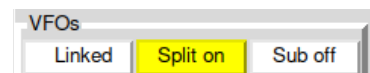
Tx/Rx button at the lower right of the app has two features. If selected, the button warns that the K4 is in transmit mode, and at it turns yellow. Note that pressing this button does not actually put out any power. It is the equivalent of using a foot switch or pressing the PTT on the mic without actually speaking. One could use it to suppress QSK in CW mode but there is very little utility in that. QSK parameters are adjustable and the results are automatic. The button also turns yellow when sending CW as the transmitter is keyed. **Pressing the space bar will toggle T/R** if the focus is on the main window.



Audio output selection is labeled “Audio Mix”. Available selection are A, B, or A+B. If other selections are specified via macros, these will be reflected in the pull-down, but they are not directly selectable from K4 Companion. XIT is selectable with the next button in this row and controlled in the RIT digit box. The box and the button turn blue for XIT and green for RIT. Both RIT and XIT frequency delta can be selected directly or by scrolling with a mouse, touchpad, or keyboard arrows. Transmit power can be selected either with pull-down presets or by direct entry.

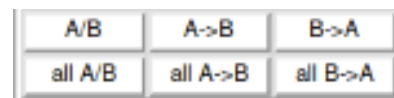
In the next row of buttons the options are:

- Linked VFOs or Unlinked. Elecraft calls this “Band Independence”
- Split on, Split off. This button lights up yellow as a secondary warning for the change of transmit VFO, and VFO B turns light blue to warn that it is now active for transmit
- Sub activates the K4 sub-receiver. This can be used either with Split mode to move the transmit VFO to B, or without. The main and sub-receivers have separate volume slider controls, located below the VFOs and meters.



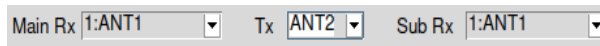
The middle row of the VFO section is self-explanatory.

- Swap VFOs A/B (may be activated repeatedly maintain different modes, for example)
- VFO A to B
- VFO B to A



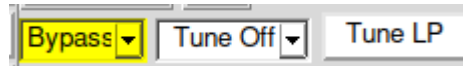
The final row “All” swaps or copies all parameters, equivalent to a “double tap” of the buttons on the K4.

Antenna selection is now available, since version 1.2.7. These combinations mirror the selections available on the K4, and various combinations can be set in Macros (see page 21.)



ATU controls function via pull-down:

- Auto = ATU active, or Bypass
- Tune = normal ATU function = single tap on the K4
- TUNE = extended tune = double tap on the K4.



The “Tune” pull-down selects two states:

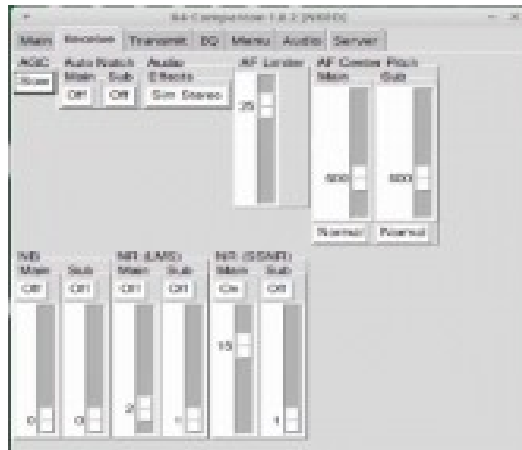
- Tune produces a carrier at the full (selected) power carrier
- Tune LP produces a low power carrier at 5 watts

The remaining button shows the “Tune” state that is active.

To EXIT K4 Companion, use the window ‘X’ at the top of the frame. Location will depend on your operating system. For Windows and Linux this is typically found at the upper right. On MacOS, it is usually on the left.

This completes the tour of the main tab of K4 Companion.

Receive Tab

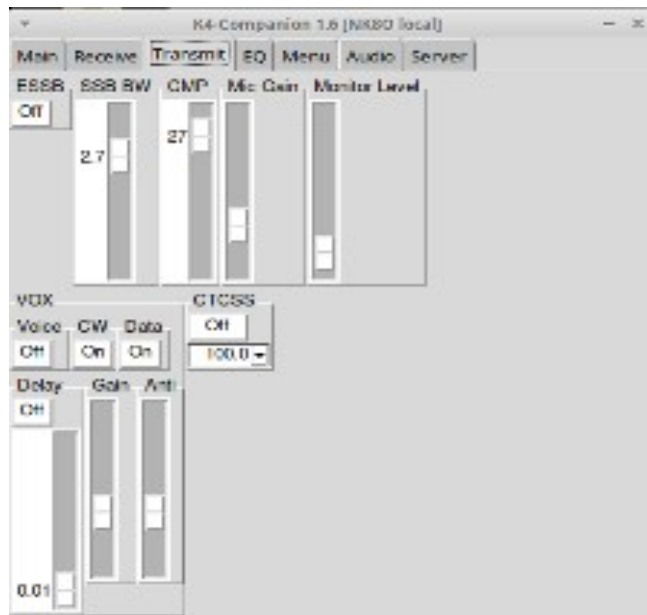


Controls for AGC, Auto Notch (SSB) and Audio Effects are in the top row. The function of each is explained in the K4 User Manual. Options for Audio effects include

- Simulated Stereo – introduces a slight delay in L – R audio to reduce listening fatigue
- Pitch map – moves audio from left to right as the deviation from the center frequency decreases or increases
- Off – removes all tuning and listening effects

Note that the AF limiter only functions if the AGC is “Off”. This prevents extremely loud signals from blasting through at full-volume if the AGC is absent. AF Center Pitch functions by mode, and returns to the last user settings when changing modes. Noise Blanking, Noise Reduction, and SSNR are explained in the K4 user manual. Recommended settings vary by band conditions and personal preferences.

Transmit Tab



- Transmit controls include ESSB (wide-band, high fidelity SSB). When selected, the SSB bandwidth varies between 3 kHz and 4.5 kHz
- SSB bandwidth for standard or CESSB (Controlled Envelope SSB) ranges from 2.4 to 2.8 kHz
- CMP, when set > 0, specifies the level of transmit audio compression when > 0, automatically enables CESSB
- Mic Gain – set to optimize SSB drive
- Monitor level – useful primarily if using K4 Companion for ancillary control of the radio while operating in the shack. No audio is returned when operating remotely. This control is independent of Sidetone level on the Main tab.
- VOX on/off with sliders to select parameters
- CTCSS – sets access PL™ tones for FM repeater operation

EQ Tab

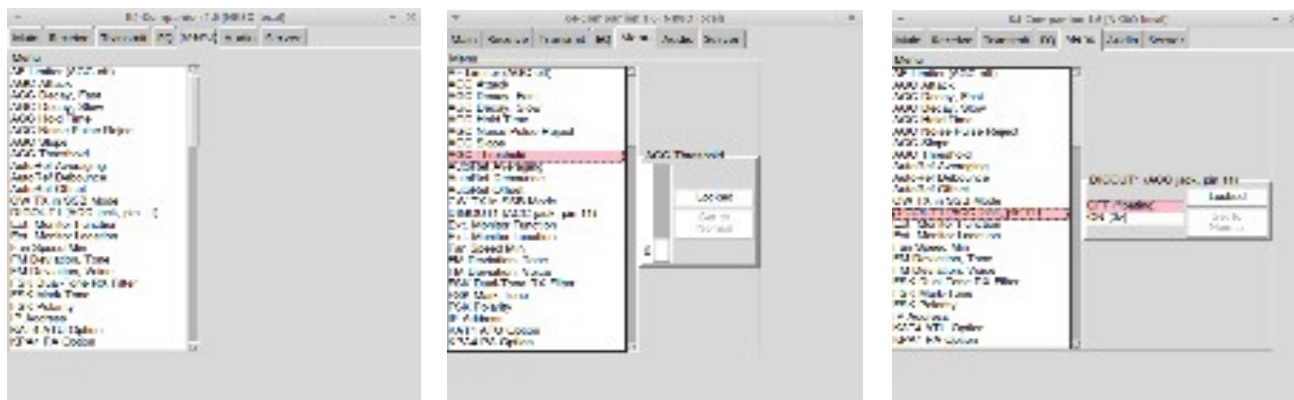


Transmit and Receive equalization can be enabled or disabled, and be set by mode. These are personal preferences. When enabled **the settings on the K4 are altered**. The K4 settings do not revert to previous settings if the Equalizers on K4 Companion are disabled.

Please refer to K4 forums or the K4 User Manual for recommendations.

Menu Tab

The Menu tab reproduces **all** of the menu functions of the K4 menu functions. These functions are covered in the user manual. The configuration menu is a long, scroll-able list. When an item is selected, it appears in pink. Then a secondary widget opens that offers the menu choices. Some of these are simple YES/NO selections, others present a list, and some have a slider to select a value for the menu item.



The images show a few of the selections possible through the menu system. Please be sure to read the K4 user manual carefully. Changing menu settings without due consideration can have unexpected results.

Audio Tab



The Audio tab currently contains four settings:

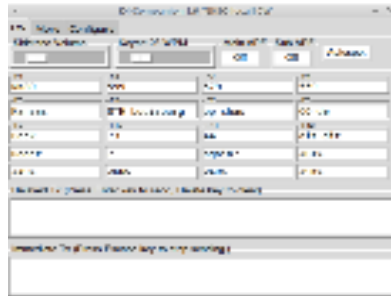
- Current Input Device
- Current Output Device
- Control K4 Audio Settings?
- Receive Latency

Typically the Input/Output settings will be correct when K4 Companion opens and they need not be changed. These settings vary by operating system. Audio connections are generally set to 'default' in a Linux installation but the choices may vary by OS 'flavor', and sound systems installed. In Windows there are a variety of settings, but often the choices mention 'Speaker' and 'Microphone.' Mac users should consistently see consistent settings that mention 'Speaker' and 'Microphone' since there is less variability in the hardware.

Control K4 Audio Settings? If you are using K4 Companion to control the radio while sitting in the shack, this can be set to Yes. Otherwise for remote operations, set it to No. See Notes on Operation below. Once this tab is configured, the settings are saved automatically. No changes are needed unless something in your operational configuration changes. The equalizer settings do change on the radio, regardless of the settings selection in the Audio tab. Frequent remote users may want to set the radio's audio to zero, and 'Control K4 Audio Settings?' to 'No'. In this way, the radio will be silent during remote operations.

The Receive Latency setting is useful for remote connections where there is considerable jitter on the network path. The lowest setting is usually acceptable in the home network setting. A very high setting could be needed for Internet connections with limited upload speeds and higher variability. Note that increasing this setting results in a slight delay in the audio that is returned to allow sufficient time for audio packets to re-assemble in the proper order. The delay becomes more prominent as the value is increased.

CW Window



K4 Companion has a CW functionality via keyboard. There are a number user configurable macros that are easily set. These can be used sequentially in any combination. The typing buffer holds any text that is set until the user hits 'enter'. The text will be sent and shown below in the Tx box. A cursor follows the text that is being sent. The sidetone will also reproduce each character audibly, unless the Sidetone level is set to zero. Tabs now divide some of the functions to prevent clutter of the interface. Audio peaking filter selection and Autospot are buttons in the CW tab. No spot tone is available remotely. Keyer speed and sidetone volume controls are on the upper left. The APF – Audio Peaking Filter – is available in the top row of the controls.

Several combination items require special characters to send correctly, but they have designations common to Elecraft. For example, 'AR' is sent with '+' and 'SK' is sent with '*'. Other prosigns include '(' for **KN**, '=' for **BT**, '%' for **AS**, '_' for **AA** (used in sending message traffic) and finally '!' for **VE**. To label these items, use the "&" as the divider to differentiate the label from what is actually sent. The label can be anything you like that tells you what is being sent. In the instance above, 'Kansas' sends 'KS'. The special characters 'AR' and 'SK' are shown as below. Symbols that will not transmit on the K4 are excluded and a brief warning appears if any is entered in the text box.

Editing the buttons is accomplished with the 'Configure' tab. The ampersand (&) is used between the label and what is actually sent. To finish editing, click 'Save'. Changes can be made at any time. In the default configuration there are 20 macro buttons. The number of rows, columns, and other configuration can be edited to taste in the 'Configure' mode, but there is no harm in leaving the default configuration. One might use one row for CWT, another for hunting parks, and another for chasing DX. There is no distinction between upper case and lower case letters when sent. Brackets <> can be used in button descriptors but not to transmit. These characters are disabled on the transmit side of the buttons



Some of the less obvious Elecraft CW designation are installed by default on the primary setup. These can be moved, changed, deleted, or edited to suit your preferences.

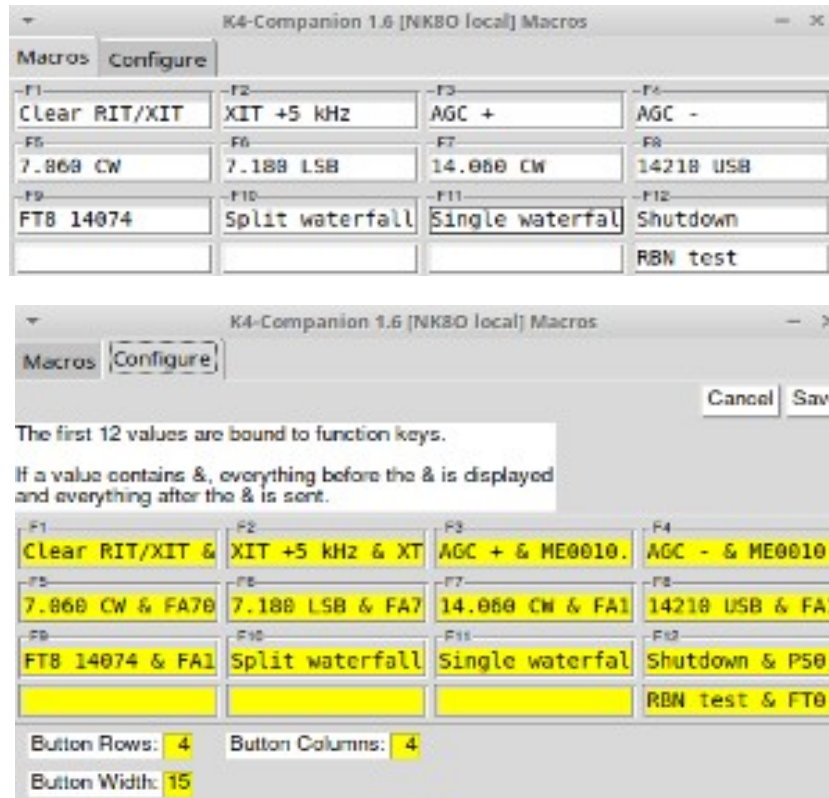
Finally, sending can be quickly canceled by hitting 'Esc' – escape.

Parameters such as CW offset/pitch are set in the "More" tab. QSK settings and CW decoder are also controlled here. The third line of the CW window does not appear unless decoding is enabled.



If the sidetone is absent despite the level being turned up, check in the Menu tab of the Main window. "TX Monitor Level, Remote" must be set to zero. If it is set to -1, there will be no sidetone heard on transmit.

Macros Window



Macros add great flexibility and control to the K4. The Macros window adds the capability of adding creating from one (1) to two-hundred (200) macro buttons. The first 12 are bound to the function keys. These can be activated if the window focus is either in the Main window or the Macros window.

The macros window can be opened and closed in the main window or closed by clicking the X in the corner of the window. The ampersand (&) is used to delineate the button description from the macro string, in the same way it is in the CW window. Please refer to the Elecraft K4 Programmer's Reference or various K4 user forums for more information regarding macro strings.

Debugging

No guarantee is made that this software will run on a particular machine or distributions. However, debugging information may be useful to the developer. Information for submissions is listed on github.com by opening an problem on the “Issues” tab of the K4 Companion site. To initiate debugging, use the ‘-d’ option: `./k4companion -d -d` . This will show a running list of actions by the script onscreen. To save the debugging information to a file, use `./k4companion -d >> mydebugfile 2>&1`. The debug file can be named anything you like and the file will be created if it doesn’t exist already. To append an existing log file, use `./k4companion -d >> mydebugfile 2>&1&`. The file can then be posted or e-mailed to Dale, W7DA. Information from `dmesg` may be helpful as well: `sudo dmesg` , although more likely if there is a system failure somewhere and not necessarily with K4 Companion.

Unfortunately the debugging option does not work with the compiled Windows EXE file. Report the problem in as much detail as possible to the developer, and copy any error messages that do appear. The scripting has been thoroughly tested prior to compiling the Windows binary, so it is likely that the problems will be found in any case.

Notes on Operation

Audio functions on Linux can sometimes be challenging. A working knowledge of ALSA, alsamixer, and alsactl is useful. Pavucontrol can assist with selecting the correct audio input and output. Be sure to check the Configuration tab in pavucontrol, and if audio is not working, try another combinations. These may include “Stereo Duplex,” “Play HiFi quality Music,” and “Pro Audio,” among others.

Regarding Windows: I have noted it is prone to random audio failures, including disabling the audio for unknown reasons. You might have to dig into the subsystem a bit. I recommend using the Sound App, not the control panel. Use *mmsys.cpl* to access the Sound App. Control panel is often not very enlightening and the Sound App gives more information. As always, YMMV – your mileage may vary!

An experiment proved that K4 Companion will run on WSL – Windows Subsystem for Linux, but it is not easy to set up, and there are quite a few stumbling blocks. Choppy audio is probably the greatest impediment, probably due to adding extra layers of complexity to the audio subsystems although this may be ameliorated with the recently added audio setting for ‘Buffered Output Packets’. It does run under Parallels in macOS, as well as other emulators.

Enjoy!

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This manual was produced using FOSS, Free Open Source Software

Recent Revision History

Update changelog for v1.6.2

Add support for setting SSNR noise reduction

v1.6:

Update changelog for v1.6

Update K4 Companion documentation and Windows executables to version 1.6

v1.5.8:

Update K4 Companion Windows EXE and Windows installer to v1.5.8

Update K4 Companion User manual to v1.5.8

Fix AF Center Pitch "Normal" button

Clicking the button would cause the associated slider to briefly go to its maximum value before going to the "Normal" value. Now it doesn't go to the maximum value.

Properly show/hide CW window on split mode change

Fix the release version in the FW Versions list

Update the main README file

Update the To Do list

Change the Transmit button color to match the VFO

Make audio device settings apply to all profiles

Done so the user only has to set them once, not for each profile.

tools/update_version: Handle no version passed

v1.5.7:

Update changelog for v1.5.7

Update Windows EXE and installer to version 1.5.7

Implements several feature requests and improvements

Update K4 Companion User Manual to version 1.5.7

update_version: Add minimal version validation

Add hidden reverse_mouse wheel direction option